🧭 Subarray Problem Approach Decision Tree

🔹 Is it a subarray problem?

Yes → Proceed to the next question.

🔹 What is the main condition?

1⃣ Count-based or Balance condition?

Examples:

- Equal number of 0s and 1s
- More 1s than 0s
- Elements greater than a certain value `k`

Approach:

- **Transformation:** Convert elements to +1, -1, or 0 based on the condition.
- **Prefix Sum:** Calculate running totals to identify balanced subarrays.
- **HashMap:** Store first occurrences of prefix sums to find maximum lengths.

2⃣ Maximum/Minimum Sum of Subarray?

Examples:

- Maximum subarray sum
- Minimum subarray sum

Approach:

- **Kadane's Algorithm:** Efficiently find the maximum sum subarray in linear time.

3⃣ Fixed or Variable Size Window Constraints?

Examples:

- Subarray of size `k`
- Subarray with sum ≤ `k`

- At most `k` distinct elements

Approach:

- **Sliding Window / Two Pointers:** Adjust window size dynamically to meet constraints.

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4 Need to Track Max/Min in a Window?

Examples:

- Maximum of every `k`-length subarray
- Longest subarray with absolute difference ≤ `k`

Approach:

- **Monotonic Deque:** Maintain elements in a way that allows constant-time access to max/min.

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5 Condition Hard to Check Directly, but Binary Search Possible?

Examples:

- Find the maximum length subarray with sum ≤ `k`
- Minimum size subarray with sum ≥ target

Approach:

- **Binary Search on Length:** Use binary search combined with prefix sums or sliding window techniques.

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6 None of the Above Fit?

Approach:

- **Brute Force:** For small input sizes.
- **Segment Tree / Sparse Table:** For range queries and updates.